Do you really know your locomotion policy?

Log the Jacobian: your global rank is hiding something.

Mind the Phase

Mind the Phase: Effective Rank and Representation Health in Legged Locomotion

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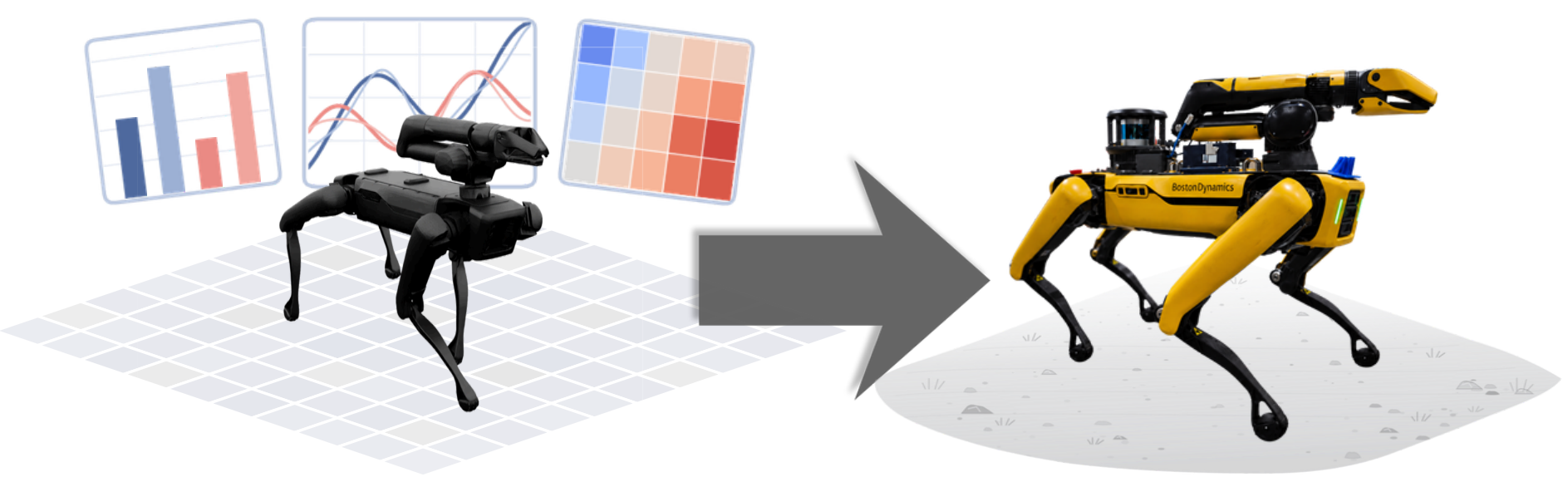
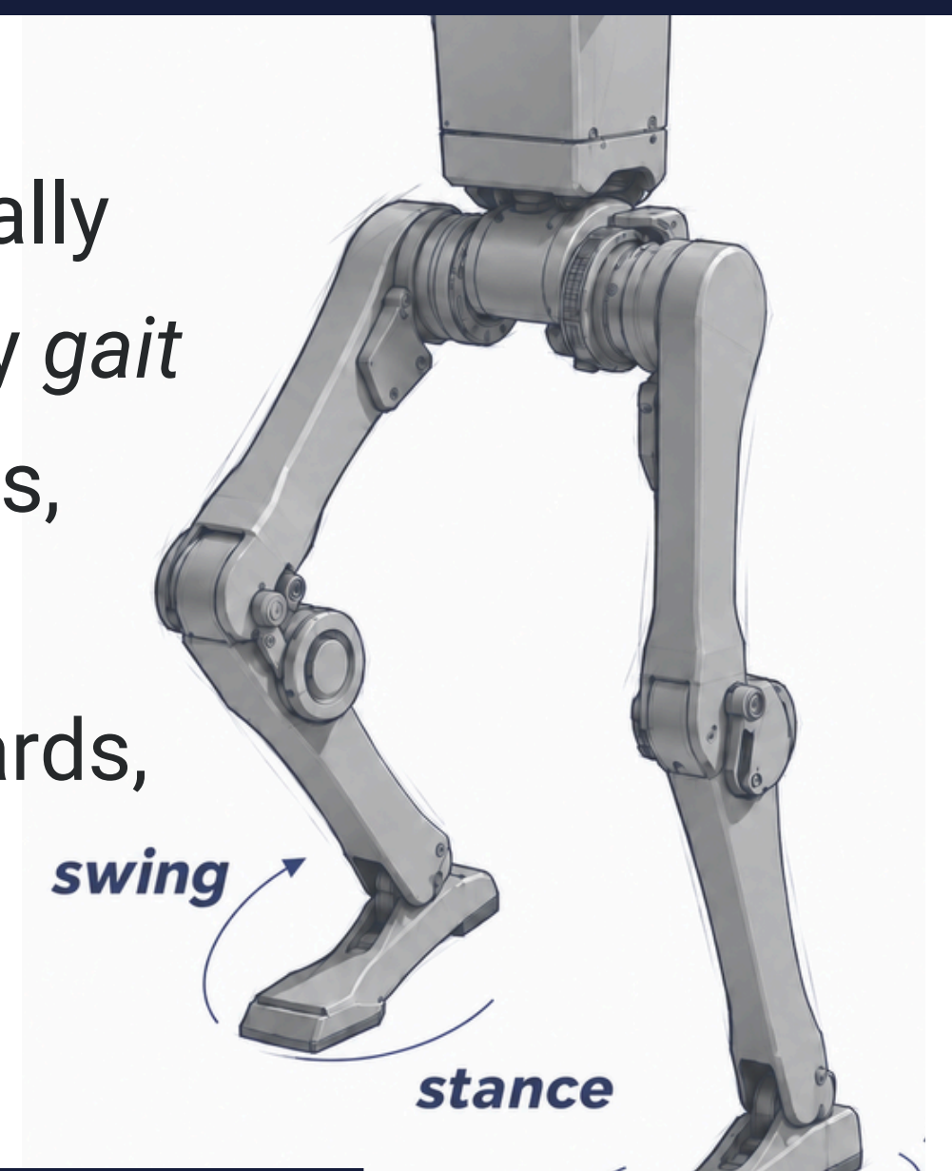
TL; DR

Log your locomotion policy's **Jacobian** rank separately for **swing**- and **stance**-dominant states. You can use the delta between them not only as a **training health** signal, but also to improve **motion smoothness** in simulation and on real hardware.



Why read our paper?



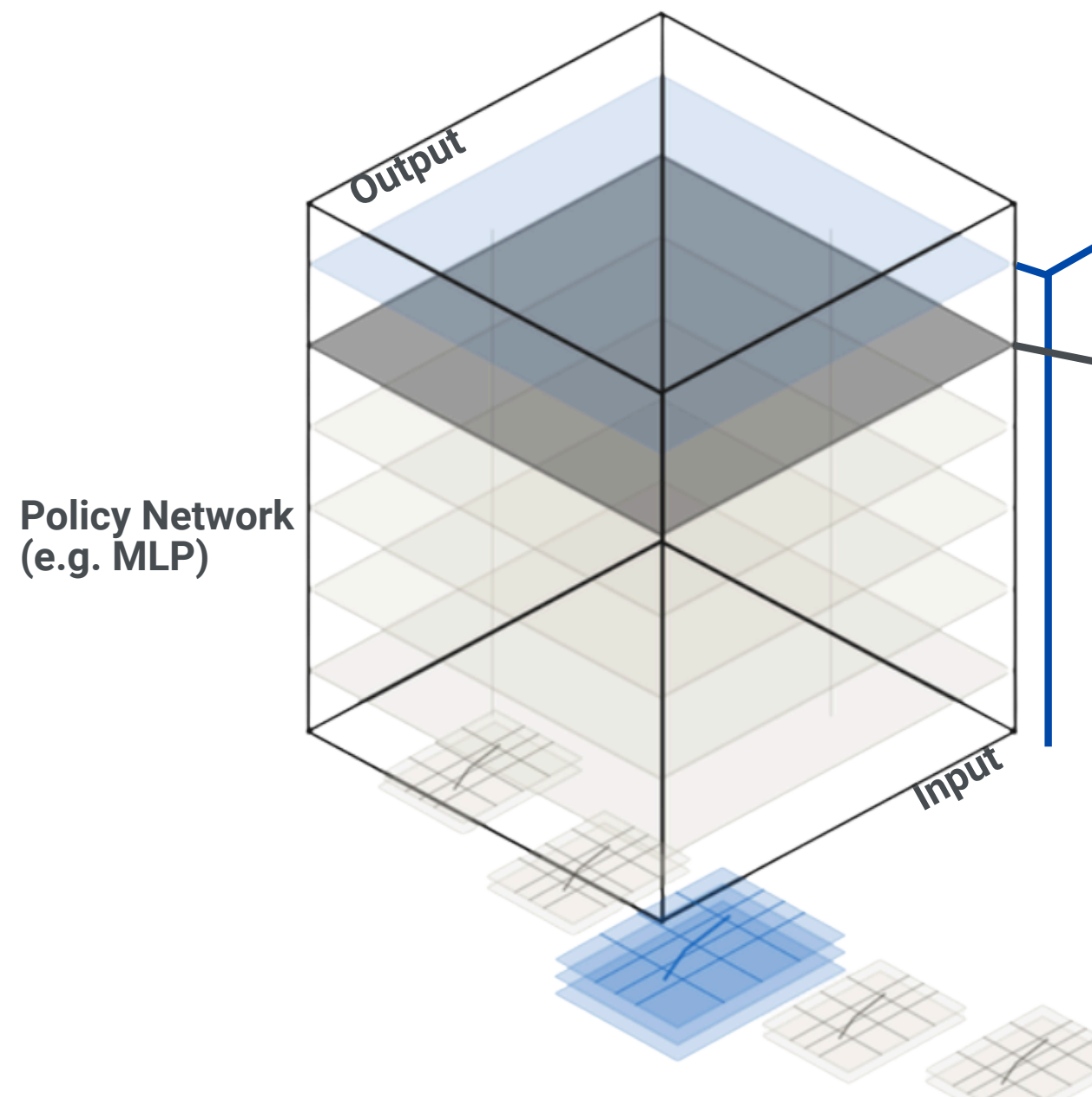
Legged locomotion has classically relied on motion coordination by *gait phases*. Modern deep RL policies, contrastingly, learn locomotion through carefully designed rewards, curriculum and environments.

But what if these policies implicitly learn to **mind the *gait* phase** during training?

Here, we look inside those policies and find that:

- effective rank offers a diagnostic of **representation health** during training time;
- **minding the *gait*-phase** reveals implicit policy structures invisible to a global rank;
- these findings accompany **smoother motion** in simulation and on real hardware.

Methods



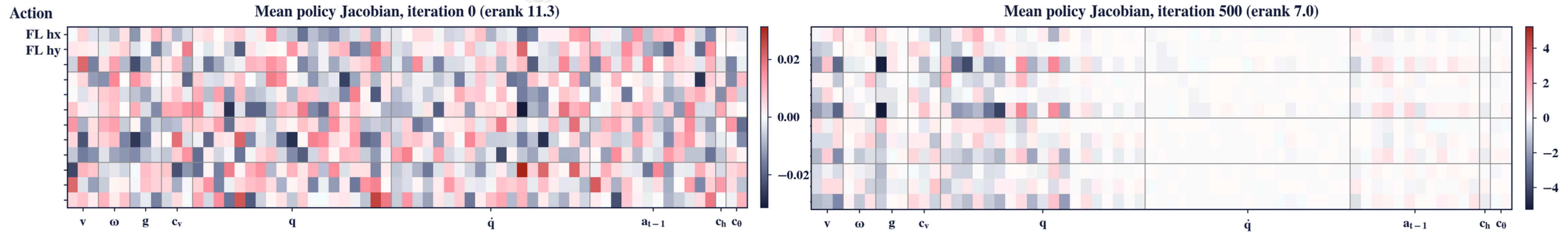
$J_\theta(x_i) = \frac{\partial \pi_\theta(x_i)}{\partial x_i}$ "Jacobian" describes how changes in observations affect actions.

$F_\theta(X)$ "Feature" describes latent representations.

Effective Rank

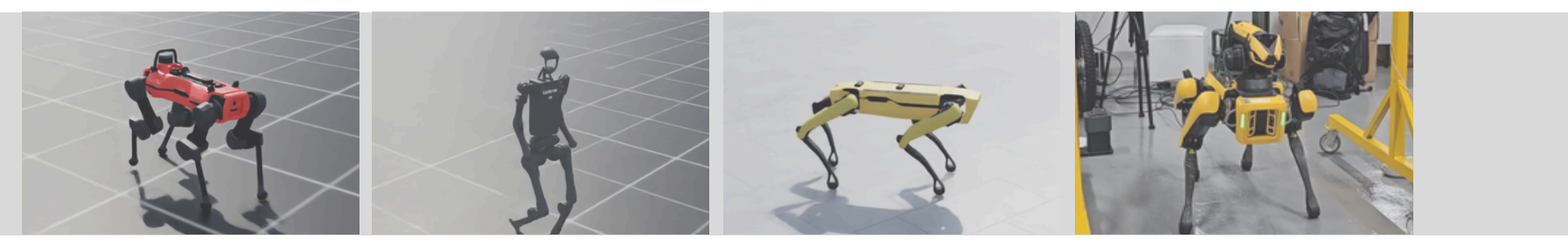
$$\mathrm{erank}(A) = \exp\left(-\sum_i p_i \log(p_i + \xi)\right),$$

number of independent directions that influence the network's output response.

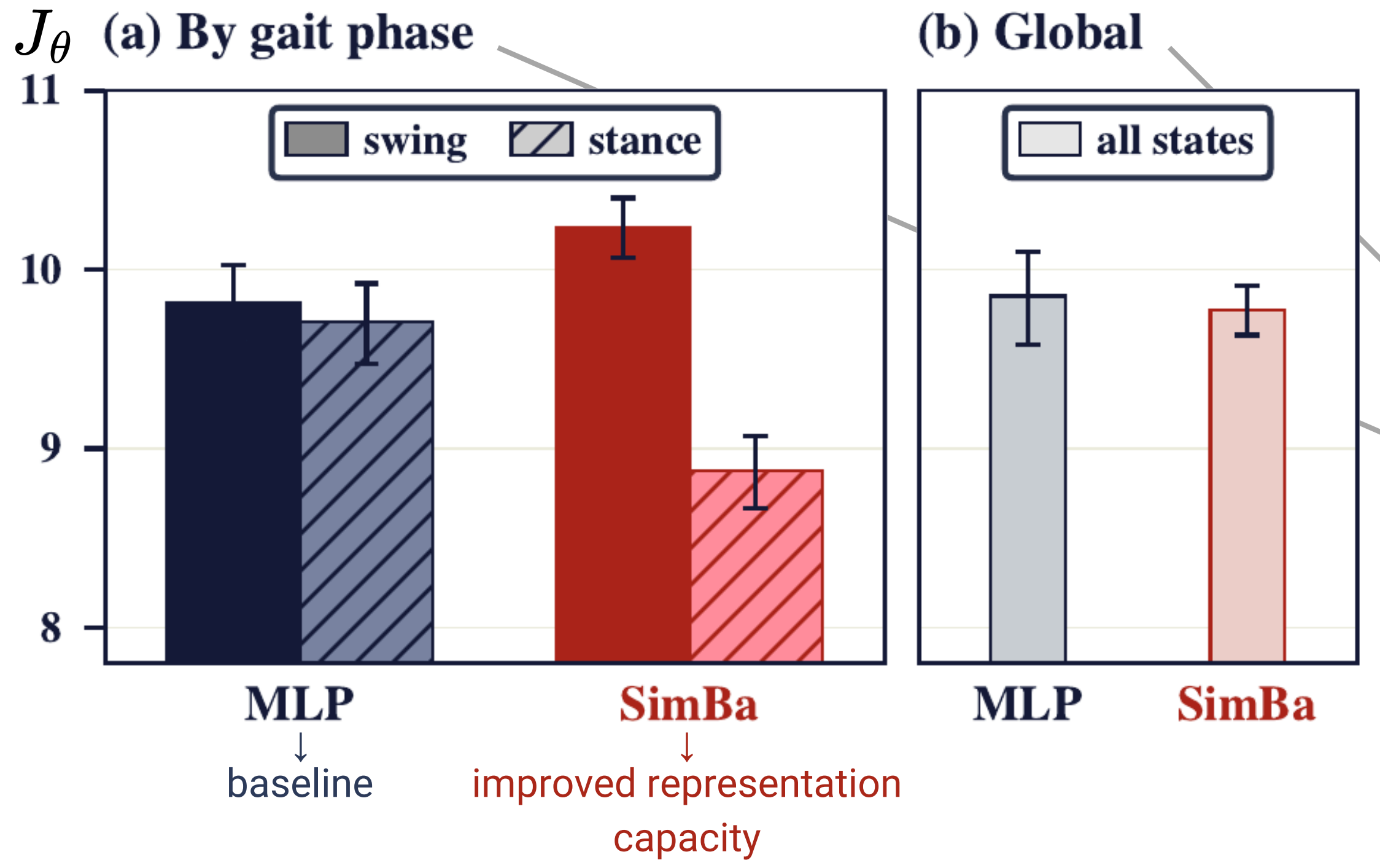


Experiments

PPO from RSL-RL within IsaacLab default tasks. 5 seeds.
Deployment on physical BDs' Spot.



1. Conditioning rank on gait phase exposes a swing-dominant split.

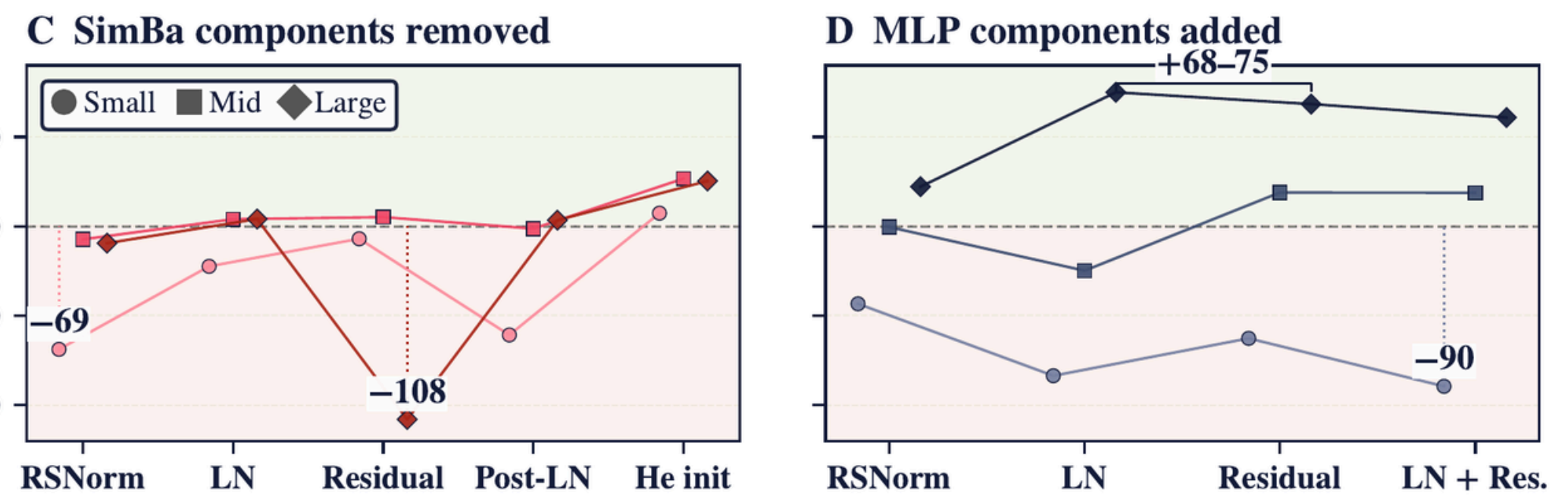


MLP
↓
baseline

SimBa
↓
improved representation capacity

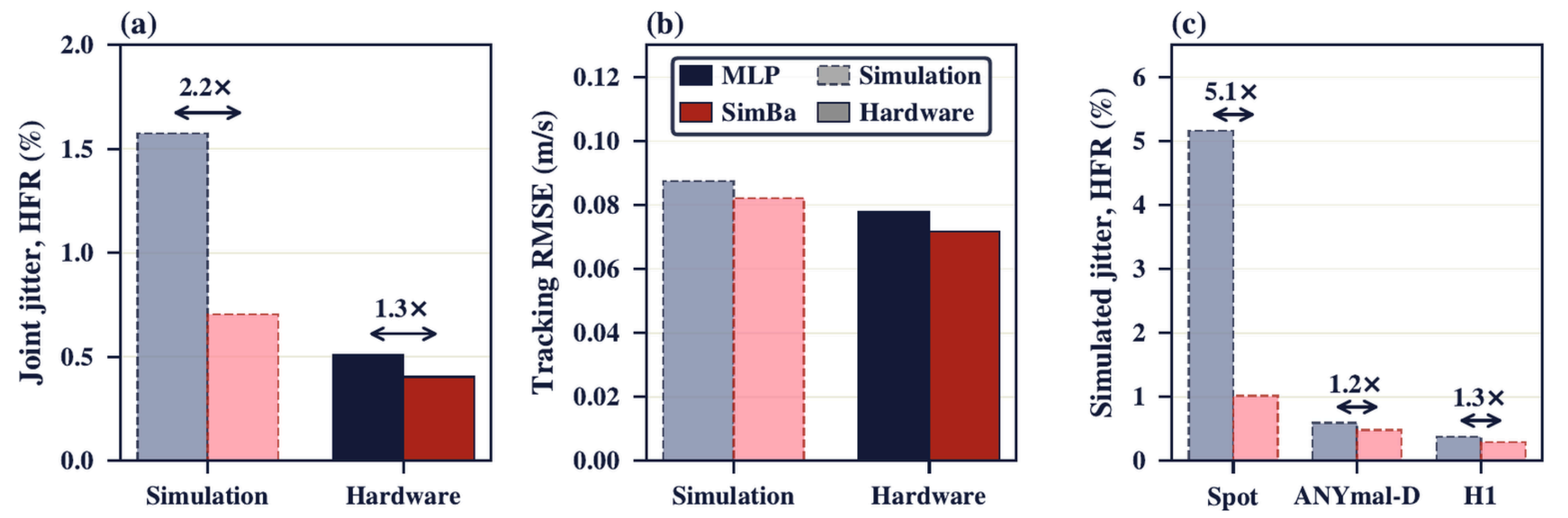
SimBa allocates more J_θ to swing than to stance-dominant states, whereas MLPs show little difference between phases.

- **Global Rank:** No signal.
- **Phase-conditioned:** Clear split when allocation is healthier.

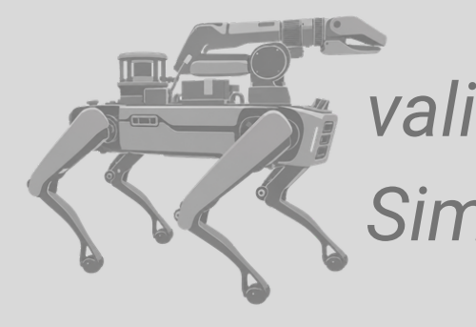


Why SimBa? Component ablations show that no single addition consistently reproduces its performance. The benefits of normalization and residual connections depend on network size.

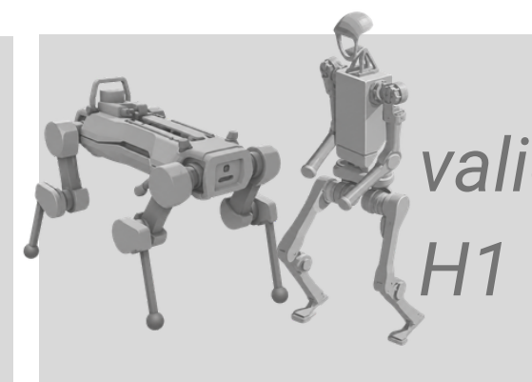
2. Higher rank policies deploy more smoothly from simulation onto hardware.



2x *less joint jitter in BDs' Spot*

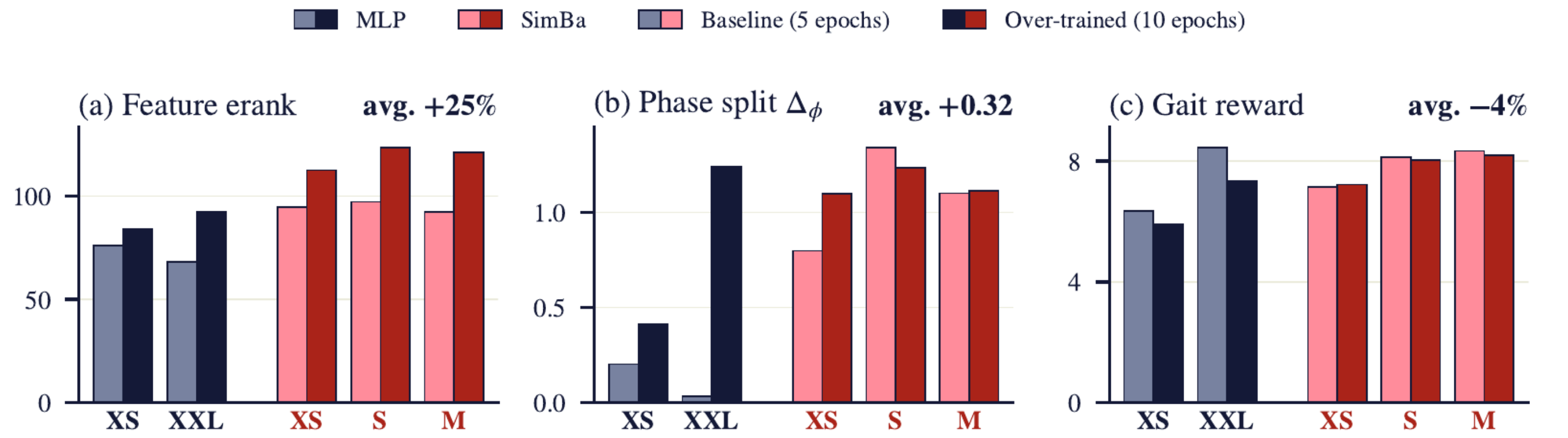


validated for Sim-to-real



validated for H1 and ANYmal in simulation

3. Higher rank is not always better.



*Over-training PPO into a forced collapse: F_θ **artificially inflates***

Effective rank therefore shouldn't be naively maximized, but used to analyze and understand representation health.

Practical Recommendations

- Log policy Jacobian effective rank, grouped by support condition (number of feet in contact with the ground);
- Record how often each support condition occurs for filtering;
- Read rank beside reward, command-tracking error and joint-motion smoothness;
- Don't maximize rank naively.

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